



PREDICTIVE ANALYTICS FOR EMERGENCY DEPARTMENT (ED) MEDICINE AND EQUIPMENT DEMAND FORECASTING USING PATIENT FLOW TRENDS AND HISTORICAL USAGE DATA

Group Members:
Hamza Nadeem (22L-6462)
Dania Baig (22L-6484)

FYP Supervisor:
Dr. Akbar Azam



INTRODUCTION

- ED environments are unpredictable.
- Hospitals still use static/manual inventory systems
- Align with the SDG of Good Health and Wellbeing
- **Result:** Shortages, overstocking, wasted budget, and risks to patient safety.
- **Solution:** a predictive analytics framework that links ED patient-flow trends with historical usage to forecast demand.
- **Tech stack:** Python for modeling, Power BI for real-time operational insights.
- **Outcome:** proactive, data-driven inventory planning instead of reactive stocking.



PROBLEM STATEMENT

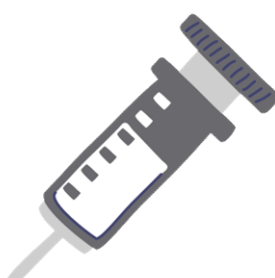
- EDs struggle to forecast medicine/equipment usage due to fluctuating demand.
- Understocking → delays, safety risks.
- Overstocking → waste, budget loss.
- Current systems ignore historical patterns + patient-flow data.

OBJECTIVE

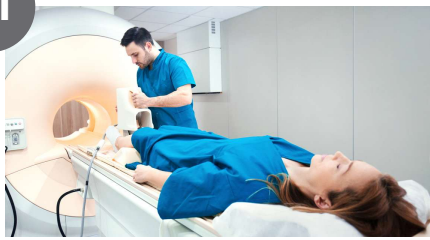
1. Build a predictive model to forecast ED item-level demand using:
 - Historical consumption
 - Patient-flow patterns
 - Machine learning/time-series techniques
1. Support optimal stocking and reduce shortages/waste.



RELEVANCE AND BUSINESS IMPACT



01



Relevance

- Enhances hospital readiness
- Reduces waste & expirations
- Improves budgeting & resource planning

02



Business Impact

- Shifts procurement from reactive → proactive
- Minimizes emergency orders
- Improves resource utilization & supply-chain performance

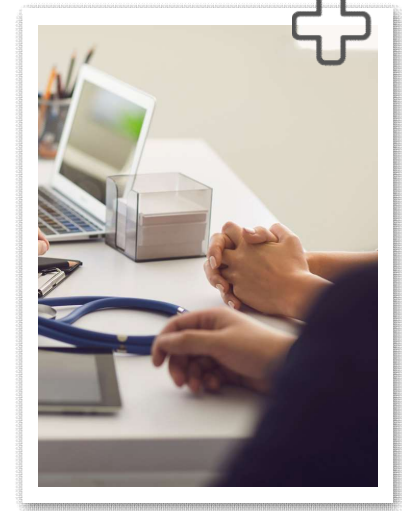
SIGNIFICANCE OF ED ML FORECASTING

Problems from poor forecasting:

- Critical shortages
- Overstocking & expiration
- Delayed patient care
- Financial inefficiency

Predictive forecasting enables:

- Better readiness
- Waste reduction
- Smarter procurement planning
- Stronger support for clinical workflows



LITERATURE REVIEW

This covers four major streams:

- Predictive analytics in ED operations
- Patient-flow trends in EDs
- Historical medicine & equipment usage
- ED medicine/equipment demand forecasting



PREDICTIVE ANALYTICS IN ED OPERATIONS: KEY INSIGHTS

- ED operations show strong temporal cycles (hourly, daily, seasonal).
- Contextual variables (weather, outbreaks, holidays) improve accuracy.
- Predictive models reduce crowding, waiting times, LWBS rates.
- Greatest impact when models are embedded into operational tools.

KEY METHODOLOGIES

- Time series (ARIMA/SARIMA)
- ML (XGBoost)
- Deep Learning (LSTM)

RESEARCH GAP

- Focus on arrivals & LOS, not inventory demand
- Mostly single-site data
- Volume-based outputs; no resource linkage
- Limited sustainability

FILLING THE GAPS

- Links arrival and acuity forecasts directly to item-level medicine/equipment demand.
- Integrates forecasting outputs into a Power BI inventory decision-support dashboard, bridging operations and procurement.
- Builds an ED-specific forecasting pipeline using real hospital usage data.
- Maps patient-flow patterns to resource consumption, providing resource-linked forecasts.
- Includes model evaluation and drift monitoring to address long-term sustainability.

PATIENT FLOW: KEY FINDINGS

- Clear intra-day, weekly, and seasonal patterns.
- High-acuity patients consume more resources.
- Flow impacted by boarding delays, downstream bottlenecks.
- External events (weather, epidemics) cause spikes.
- Variability, not averages, drives operational stress.

KEY METHODOLOGIES

- Trend decomposition, regression
- ARIMA/SARIMA, Prophet

RESEARCH GAP

- No explicit link to item-level consumption
- Aggregated flow models ignore clinical pathways
- Weak multi-source data integration
- Minimal evaluation of inventory impact
- Non-stationarity often overlooked

FILLING THE GAPS

- Connects patient-flow trends to per-item consumption using historical ED usage data.
- Incorporates triage levels, diagnosis groups, and acuity mix to represent true clinical pathways.
- Combines ED arrival data, usage data, and temporal features into a unified modeling pipeline.
- Evaluates how flow-driven demand forecasts can reduce shortages and improve inventory accuracy.
- Uses seasonal features, variability metrics, and cross-validation to account for non-stationary ED patterns.

HISTORICAL EQUIPMENT AND MEDICINE USAGE: KEY FINDINGS

- Strong correlation between patient volume & consumption.
- High-acuity/diagnosis groups have higher per-visit usage.
- Seasonal patterns follow disease trends.
- ABC analysis → few items = most cost/clinical risk.

KEY METHODOLOGIES

- Trend analysis, safety stock/reorder point models
- ARIMA
- ML forecasting models
- ABC Analysis

RESEARCH GAP

- Little ED-specific modeling
- Weak linkage to patient flow
- Data quality inconsistencies
- Focus on cost over clinical criticality
- Lack of real-time analytics

FILLING THE GAPS

- Builds ED-focused item-level demand models rather than hospital-wide estimates.
- Establishes a direct link between historical usage and patient-flow/acuity patterns.
- Cleans and structures usage data for forecasting, improving reliability.
- Prioritizes clinically critical items using ABC/criticality frameworks rather than cost-based metrics alone.
- Implements a forecasting dashboard that can be extended to real-time analytics in future work.

ED INVENTORY FORECASTING: KEY INSIGHTS

- Demand driven by arrivals + acuity (non-linear relationships).
- Models with clinical covariates outperform simple historical usage.
- Forecast-driven inventory improves stock-outs/wastage.
- Dashboards improve planning and decision-making.

KEY METHODOLOGIES

- Multivariate regression
- ARIMA/SARIMA
- Random Forest / XGBoost

RESEARCH GAP

- Few integrated end-to-end pipelines
- Limited item-level demand modeling
- Little consideration of uncertainty & lead times
- Minimal real-world deployment
- Weak generalizability

FILLING THE GAPS

- Develops a full end-to-end forecasting pipeline: flow → consumption → demand → dashboard.
- Creates item-level and class-level prediction models tailored for the ED.
- Incorporates variability, seasonality, and confidence intervals to reflect uncertainty.
- Deploys outputs in a Power BI dashboard for real-world operational use.
- Builds a framework that can be adapted to different hospitals with adjustments to features and calibration.

EXISTING RESEARCH



Forecasting ED patient arrivals for operational readiness



Hospital supply-chain management to prevent drug stockouts



Predictive analytics and machine learning in healthcare



Real-time dashboards and workflow visualization in hospitals

GAP



Limited research linking patient-flow forecasting-in item-level inventory prediction in EDs



Limited research linking patient-flow forecasting with item-level inventory prediction

OUR STUDY



Developing integrated forecasting models



Combining clinical and consumption patterns



Leveraging predictive analytics for ED inventory

METHODOLOGY OVERVIEW

Objective:

- Forecast medicine and equipment demand in EDs using a hybrid analytical approach.

Key Components:

- Descriptive Analytics
- Data Visualization
- Predictive Analytics (Time-series + ML + Deep Learning)
- Visual: Flowchart of methodology stages



DATA SOURCES

1. KAGGLE HOSPITAL DATASET (PILOT)

HISTORICAL ED ENCOUNTERS, ADMISSIONS, DISCHARGES

EXCEL SHEETS: NUMERICAL, CATEGORICAL, TEMPORAL VARIABLES

PURPOSE: TEST CLEANING, VISUALIZATION, FEATURE ENGINEERING

2. EVERCARE ER DATASET (FINAL)

2-3 YEARS OF ANONYMIZED ED DATA

INCLUDES PATIENT ARRIVALS, TRIAGE, MEDICINE & EQUIPMENT USAGE, BED
OCCUPANCY

MAIN SOURCE FOR FORECASTING

DATA PRIVACY & ETHICS

- Anonymization of all personal identifiers
- Compliance with GDPR/HIPAA-like standards
- Bias prevention checks across demographics
- Ethical use: Aggregate-level analysis only



DATA PREPARATION & CLEANING



Steps:

- Remove duplicates & incorrect entries
- Handle missing values (median/mode imputation)
- Correct data types & standardize categories
- Derived variables: LOS, Age, DaysOnStaff, HourOfVisit
- Referential integrity checks

DESCRIPTIVE ANALYTICS

Purpose: Understand past trends & identify ED patterns

Techniques:

- Mean, median, SD, IQR for vitals, LOS, daily visits
- Histograms, boxplots for distributions & outliers
- Correlations & crosstabs between patient acuity, service type, resource consumption

DATA VISUALIZATION

Purpose: Explore relationships & trends visually

Techniques:

- Line charts & heatmaps: daily, weekly, seasonal trends
- Bar & pie charts: patient distribution by service/discharge
- Word clouds: common ED visit reasons
- Sunburst charts: provider specialties & experience

PREDICTIVE ANALYTICS

Purpose: Forecast medicine & equipment demand

Models:

- **Time-Series:** ARIMA, Prophet → trends, seasonality, holiday effects
- **Machine Learning:** XGBoost → multivariate relationships
- **Deep Learning:** LSTM → long-term temporal patterns

STATISTICAL ANALYSIS

- **Correlation analysis:** numerical variables & inventory usage
- **Hypothesis testing:** t-tests, chi-square for categorical comparisons
- **Validates predictive variables for modeling**

FEATURE ENGINEERING

Derived Variables:

- **LOS, Age, Peak/Off-Peak visit flags**
- **Daily triage distribution**
- **Seasonal indicators (winter/summer, holidays)**

Purpose: Improve predictive accuracy of ML and LSTM models

HYBRID FORECASTING APPROACH

Rationale:

- **ED demand is dynamic & non-linear**
- **Single model insufficient → combine:**
- **Time-series → temporal patterns**
- **ML → non-linear interactions**
- **LSTM → long-term sequences**

Outcome: Accurate, actionable forecasts

DASHBOARD & VISUALIZATION

Tools: Power BI / Plotly

Outputs:

- **Daily visit trends**
- **Forecasted demand curves**
- **Inventory risk alerts**
- **Sunburst charts: provider → specialty → visits**
- **Heatmaps: peak ED hours**

PYTHON PACKAGES

Data Cleaning & EDA: pandas, numpy, matplotlib, seaborn, plotly

Time-Series Forecasting: statsmodels, fbprophet

Machine Learning: scikit-learn, xgboost

Deep Learning: tensorflow / keras

Text Analysis: wordcloud

Utilities: datetime, warnings

METHODOLOGY JUSTIFICATION

- **Descriptive & EDA → operational baseline**
- **Statistical tests → validate patterns**
- **Time-series → capture temporal trends**
- **ML → multivariate effects**
- **LSTM → long-range sequences**
- **Dashboards → communicate insights**
- **Outcome: Scalable, accurate, and clinically relevant forecasts**

BENEFITS TO STAKEHOLDERS

Hospital Administrators: proactive procurement, reduced wastage

Inventory Teams: dynamic reorder points, budgeting efficiency

Clinicians/Nurses: timely medicine & equipment availability

Hospital Leadership: evidence-based planning

SUMMARY

- **Hybrid forecasting integrates descriptive, statistical, ML, and deep learning approaches**
- **Data cleaned, anonymized, and standardized for accuracy**
- **Dashboarding enables real-time decision-making**
- **Provides a predictive, data-driven ED inventory management system**





THANK YOU

Group Members:
Hamza Nadeem (22L-6462)
Dania Baig (22L-6484)

FYP Supervisor:
Dr. Akbar Azam